

Multi-Table Sales Analysis

Northwind Orders & Order Details | SQL + Power BI

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Tools: SQL (SQLite), Power BI Desktop

Net Sales	Orders	Customers	Avg Order Value
\$1,265,793	830	89	\$1,525

1. Project Overview

Objective: Apply relational analysis end to end - combine multiple tables, validate the joins, avoid double counting, and turn the result into business insights and a dashboard.

Deliverables: SQL script, Power BI dashboard, 5 business insights (all covered in this report).

2. Dataset

Table	Rows	Grain (1 row =)	Key columns
orders	830	one order	order_id (PK), customer_id, employee_id, order_date, freight, ship_country
order_details	2,155	one product line in an order	order_id (FK), product_id, unit_price, quantity, discount

Relationship: orders (1) to order_details (many) on *order_id*. Data period: 04 Jul 1996 to 06 May 1998 (1998 is a partial year).

Limitation: the supplied files had no separate products or customers table, so analysis uses product_id and customer_id (customer country is derived from orders). Adding those two tables would only add names/categories - the join logic stays the same.

3. Method

Net Sales = unit_price x quantity x (1 - discount), calculated at order-line level in a SQL view (vw_sales_lines). A second view at order level (vw_orders_summary) was created to cross-check totals. Power BI loads these views and uses DAX measures on top.

4. Join Validation

Check	Result	Status
Duplicate order_id in orders	0	Pass
Orphan rows in order_details (no parent order)	0	Pass
Orders with no items	0	Pass
Rows after JOIN vs order_details rows	2,155 = 2,155	Pass
Total from line view vs order view	\$1,265,793.04 = \$1,265,793.04	Pass

5. Double Counting - Demonstrated

freight lives in the orders table (one value per order). After joining to order_details, every order is repeated once per product line, so summing freight on the joined data inflates it.

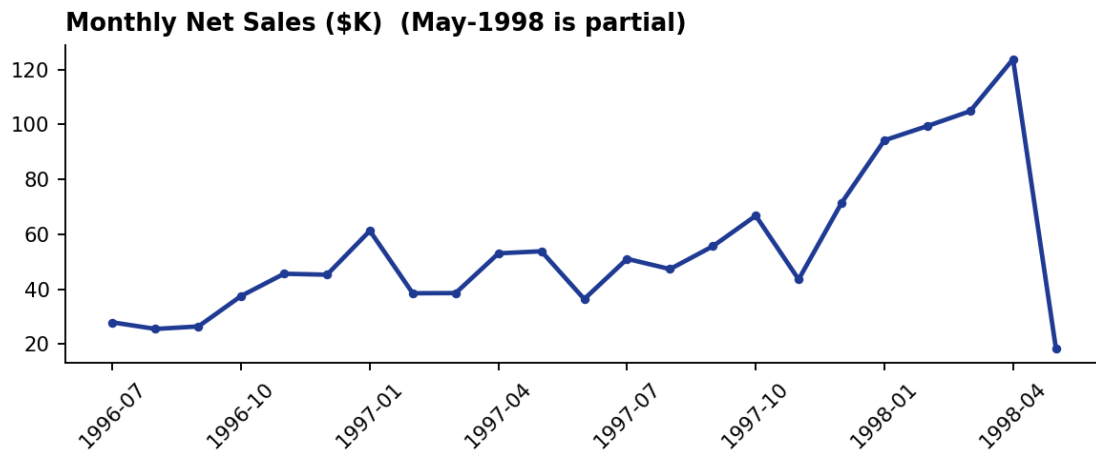
Measure	Value
True total freight (orders table)	\$64,942.69
Freight summed after JOIN (wrong)	\$207,306.10
Inflation	3.19x

Fix: aggregate to the correct grain first (one row per order) and then sum, or use measures on the orders table only. Sales amounts are safe because they are calculated per line.

6. Five Business Insights

Insight 1 - Strong growth

1997 sales were \$617K versus \$208K in 1996 (which has only Jul-Dec). Jan-Apr 1998 reached \$422K against \$191K for Jan-Apr 1997 - about **2.2x** year over year. May 1998 is partial, so the drop at the end of the trend is a data cut-off, not a decline.



Insight 2 - Sales depend on three countries

USA (\$246K, 19.4%), Germany (\$230K, 18.2%) and Austria (\$128K, 10.1%) together give **47.7%** of net sales. Brazil and France follow.

Insight 3 - A few customers matter a lot

QUICK (\$110K), ERNSH (\$105K) and SAVEA (\$104K) contribute **25.2%** of sales from only 3 of 89 customers. Losing one of them would be a visible hit.

Insight 4 - One product leads clearly

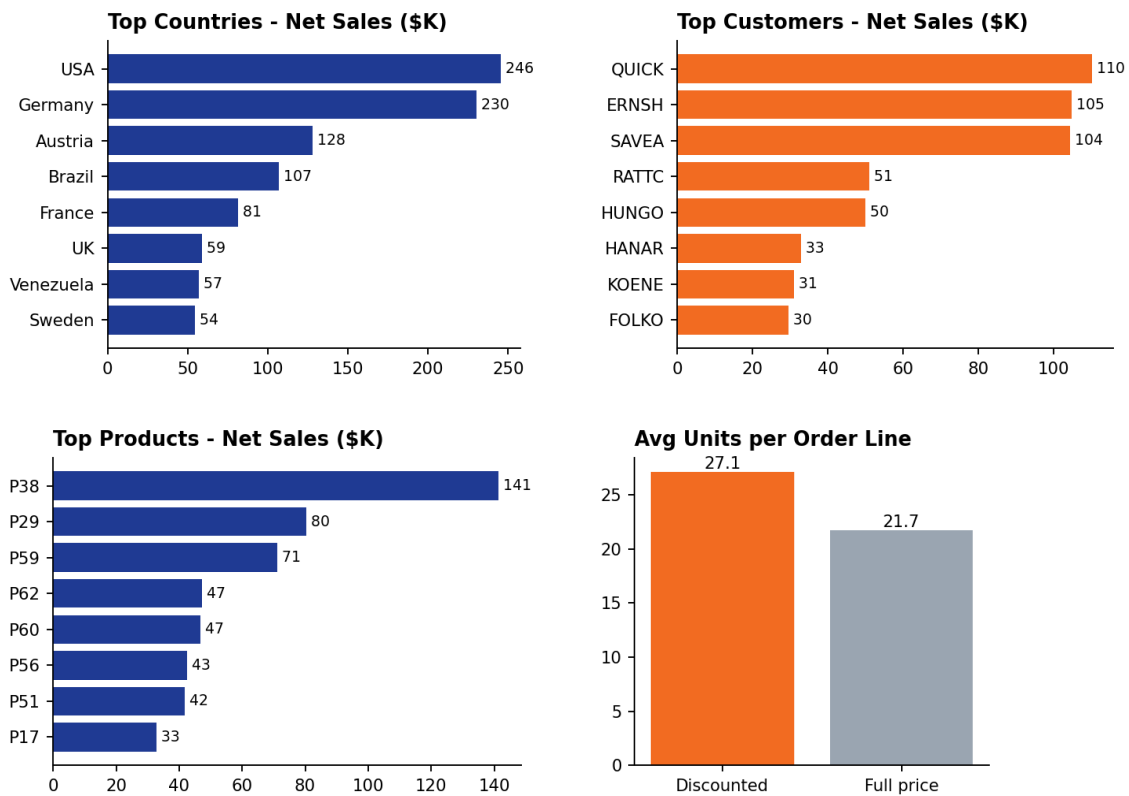
Product 38 earns \$141K (about 11% of sales) - almost double product 29 (\$80K). Product 59 sells the most volume among top earners (1,496 units) but earns half of product 38.

Insight 5 - Discounts raise volume but cost money

38.9% of lines carry a discount. Discounted lines average **27.1 units** versus 21.7 at full price (+25%), but discounts gave away **\$88.7K** (6.5% of gross sales). Full-price lines still produced the larger share of revenue (\$751K vs \$515K).

Extra: 37 of 809 shipped orders (4.6%) were delivered after the required date - a service-level point to monitor.

7. Analysis Charts



8. Power BI Dashboard

Dashboard contents: 4 KPI cards (Net Sales, Orders, Customers, Avg Order Value), monthly trend line, top countries, top customers, top products, year column chart, and slicers for Year and Country.

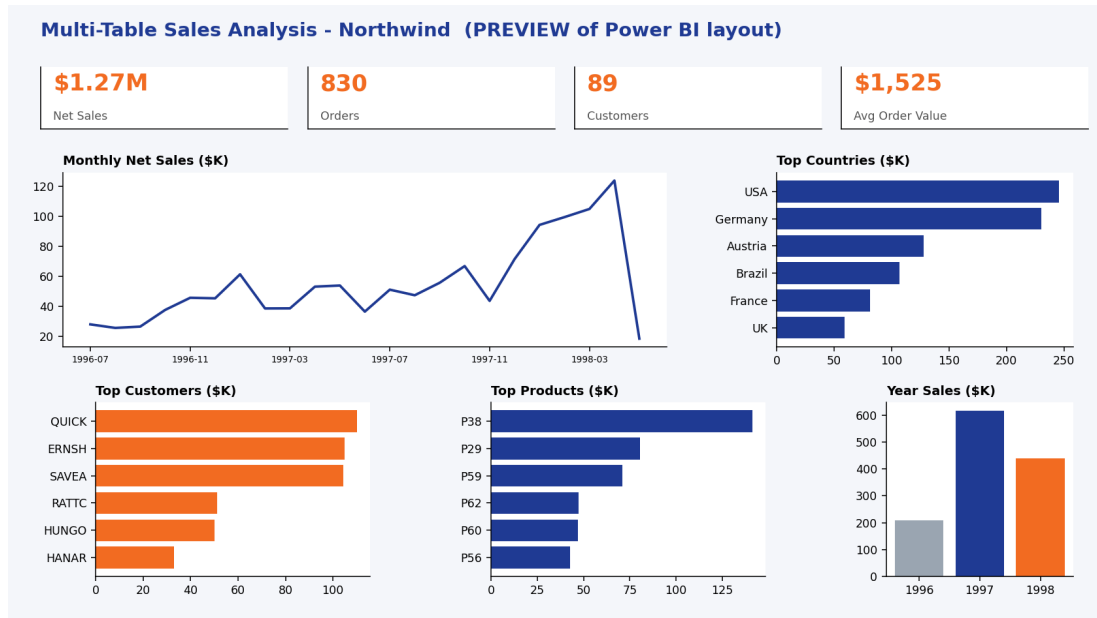


Figure: Dashboard layout reference.

9. Recommendations

1. Protect the top 3 customers with account management and reduce dependence by growing mid-sized accounts.
2. Push marketing in Brazil, France and UK, which are strong but below the top three.
3. Keep product 38 well stocked and test bundles with high-volume products.
4. Use targeted, not blanket, discounts - volume rises ~25% but 6.5% of gross sales is given away.
5. Review the 4.6% late shipments by shipper and country.

10. Interview Questions

How can joins cause double counting?

A join to a table with a finer grain (many rows per key) repeats the parent row once per child row. Any parent-level value (like freight) then gets summed multiple times. Fix it by aggregating to the correct grain before joining or by measuring on the right table.

How do you validate totals?

Compare the same total through independent paths: raw table vs joined view vs order-level view, check row counts before and after the join, look for orphan keys and duplicates, and reconcile the Power BI card to the SQL result. Here all totals matched at \$1,265,793.04.

11. Conclusion

The two tables joined cleanly with no orphans or duplicates. By computing sales at line level and validating totals in multiple ways, the analysis avoids double counting and gives reliable insights: strong growth, geographic and customer concentration, a leading product, and a discount trade-off.

Files: multi_table_sales_analysis.sql | vw_sales_lines.csv | vw_orders_summary.csv | vw_customers.csv | dashboard screenshot